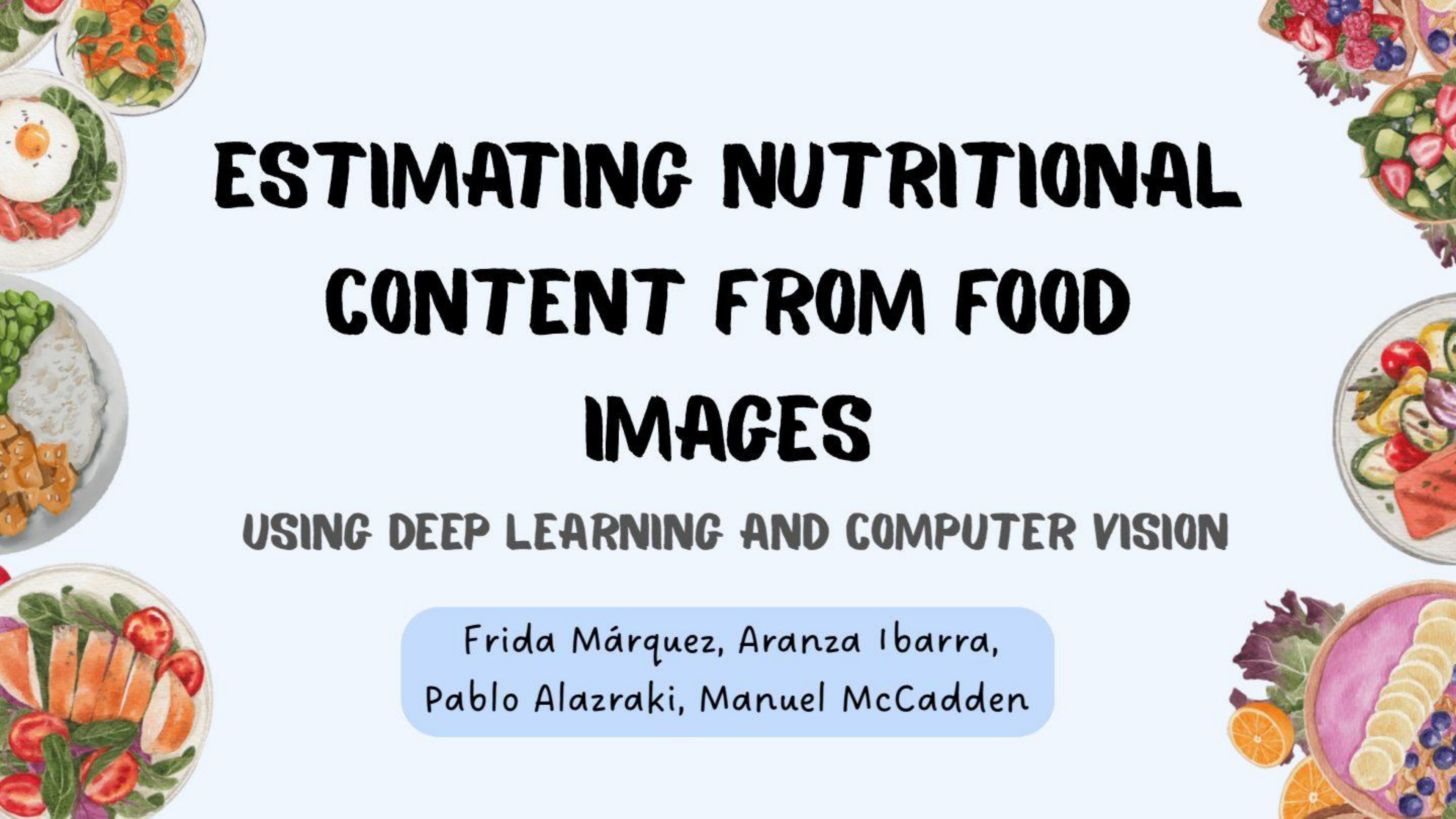


The slide features a light blue background with various food illustrations. In the top left, there are bowls of salad and a plate with a fried egg. In the bottom left, a plate of salmon and tomatoes is visible. On the right side, there are more bowls of fruit and salad. The main title is centered in a large, bold, black font.

ESTIMATING NUTRITIONAL CONTENT FROM FOOD IMAGES

USING DEEP LEARNING AND COMPUTER VISION

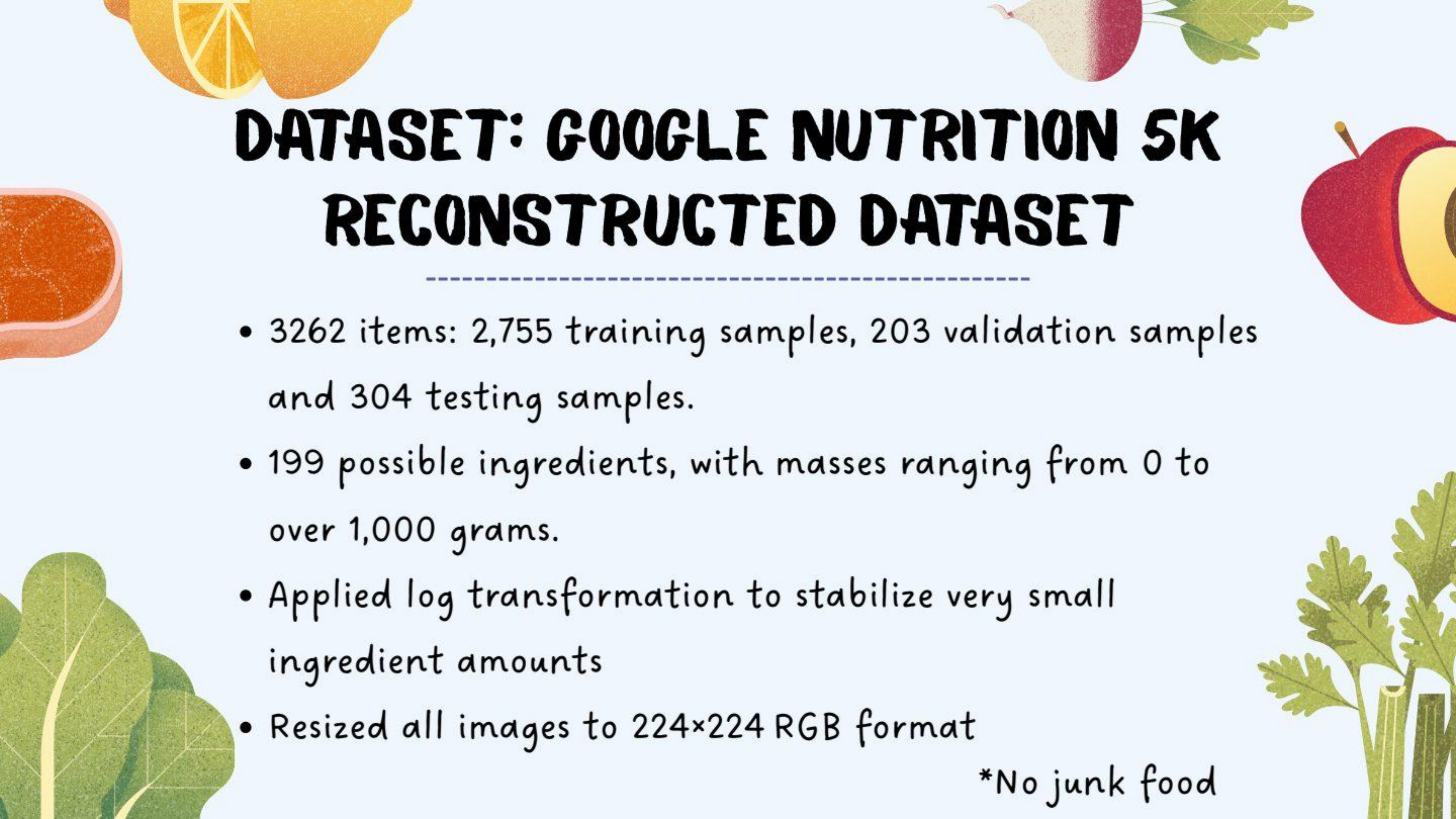
Frida Márquez, Aranza Ibarra,
Pablo Alazraki, Manuel McCadden



MOTIVATION

Living with type 1 diabetes makes estimating the carbohydrate content of meals a constant and often frustrating challenge.

By combining deep learning with nutritional datasets, this project aims to support people with diabetes in making more informed decisions about their food intake improving both convenience and glucose control.

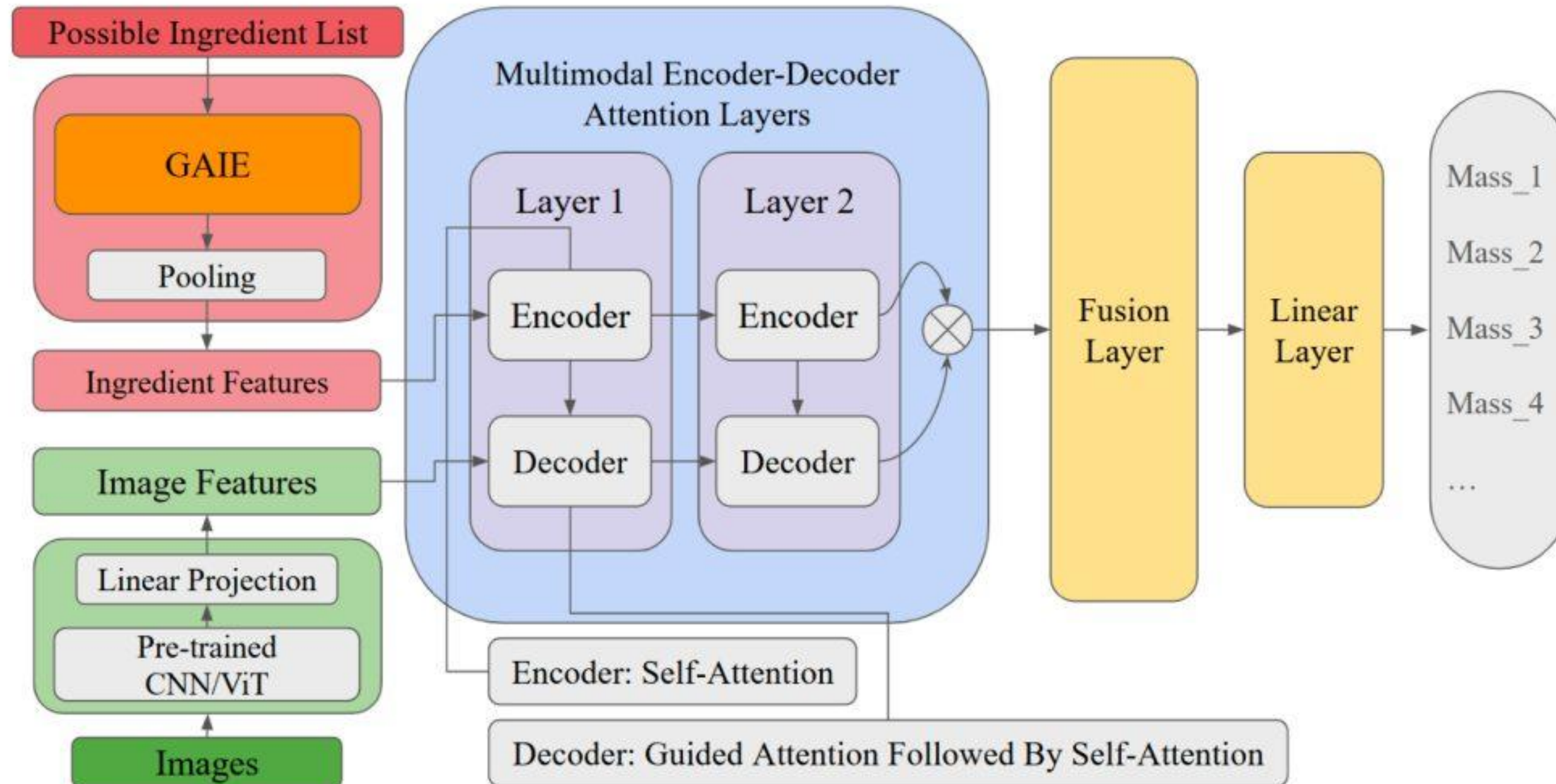


DATASET: GOOGLE NUTRITION 5K RECONSTRUCTED DATASET

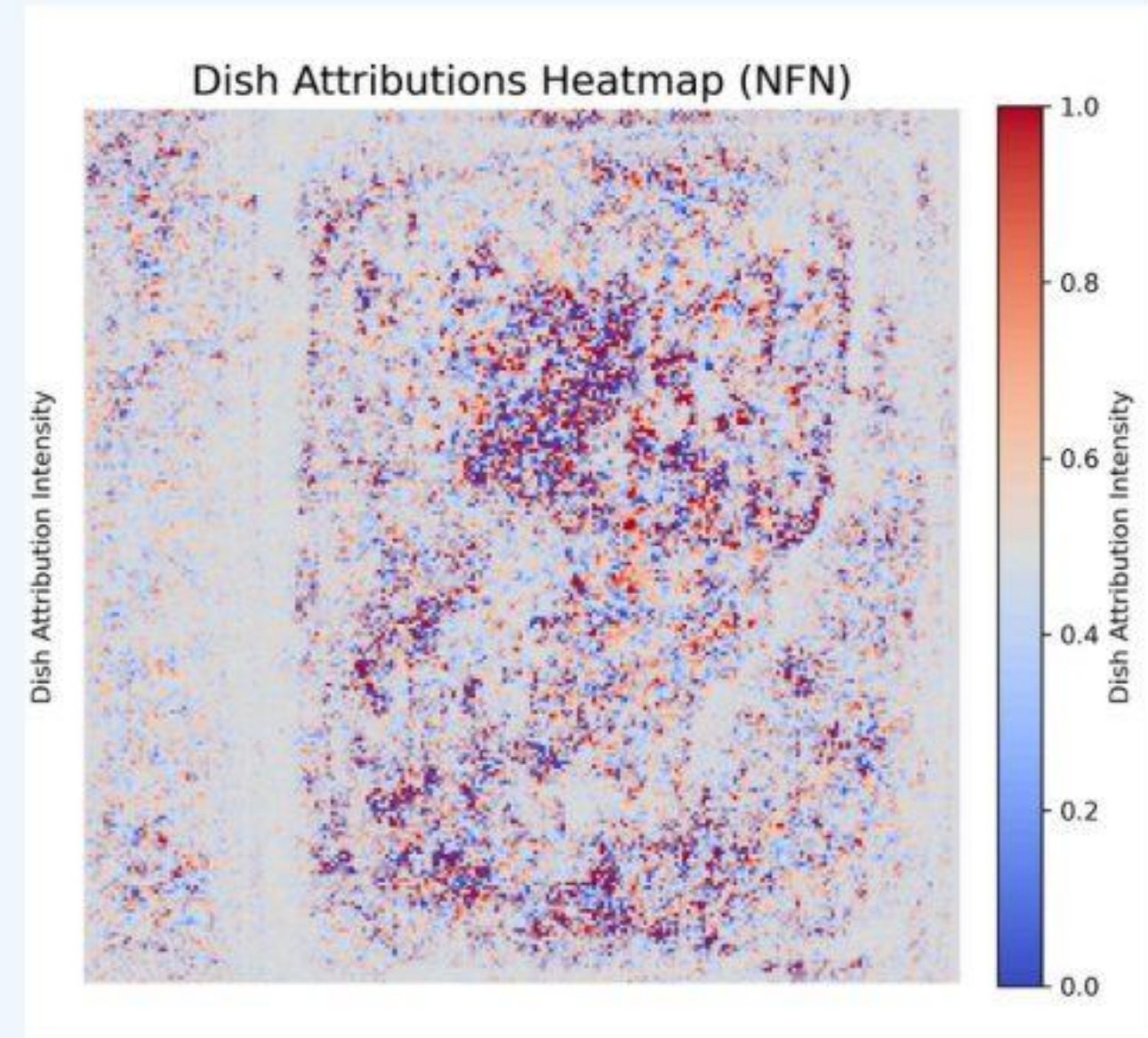
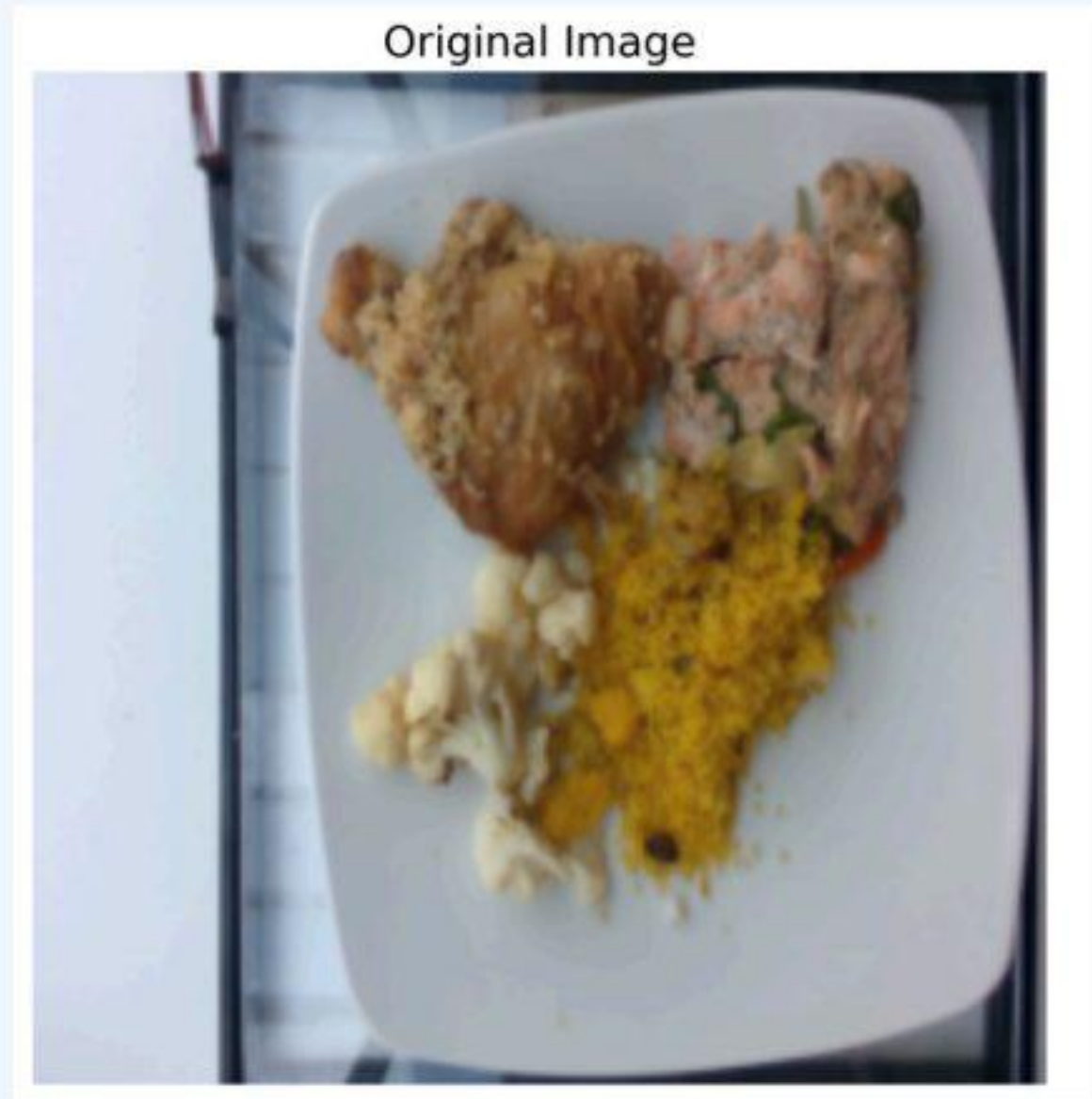
- 3262 items: 2,755 training samples, 203 validation samples and 304 testing samples.
- 199 possible ingredients, with masses ranging from 0 to over 1,000 grams.
- Applied log transformation to stabilize very small ingredient amounts
- Resized all images to 224×224 RGB format

*No junk food

BASELINE MODEL ARCHITECTURE: NUTRIFUSIONNET



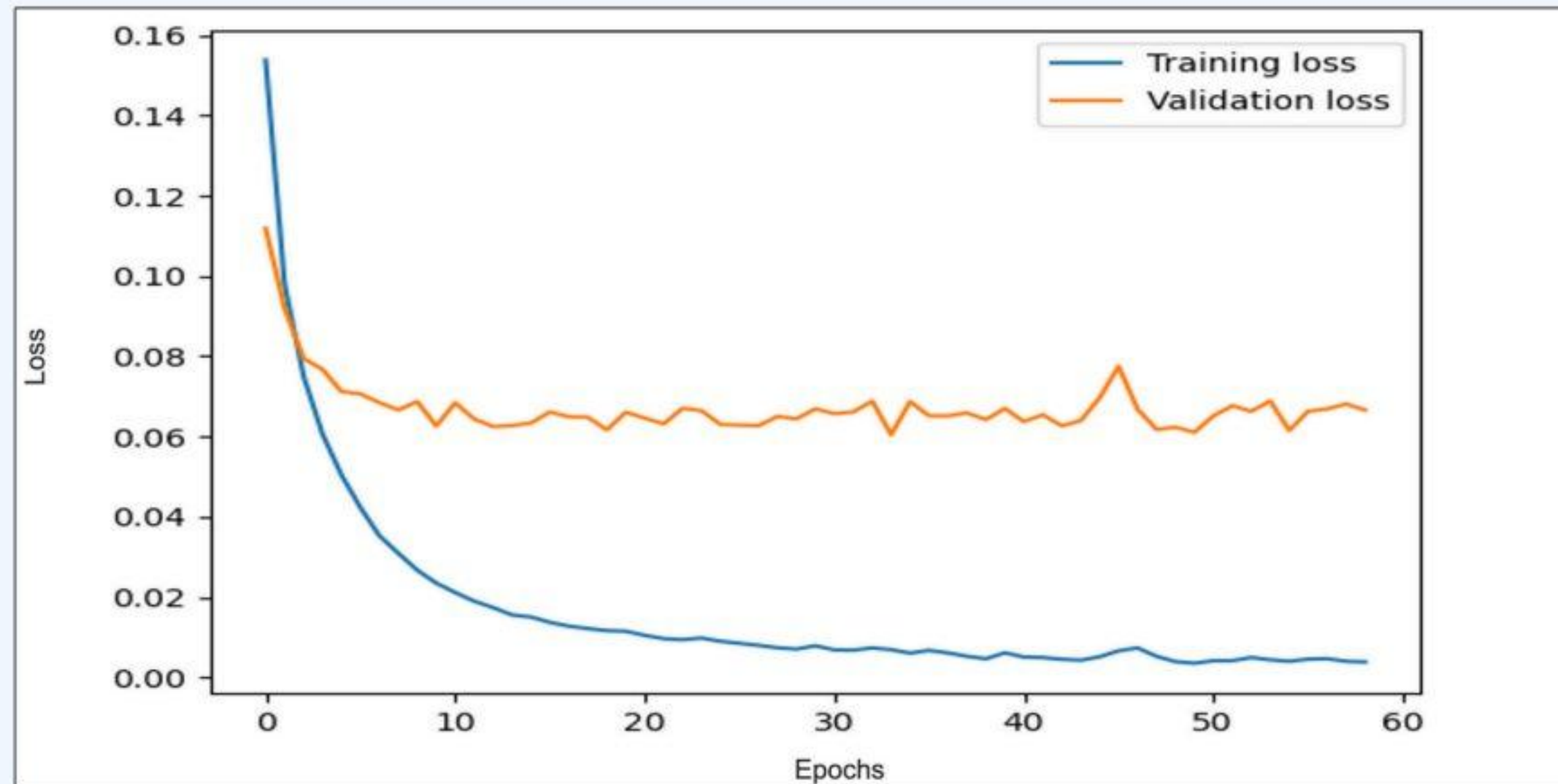
INGREDIENT CLASSIFICATION PERFORMANCE



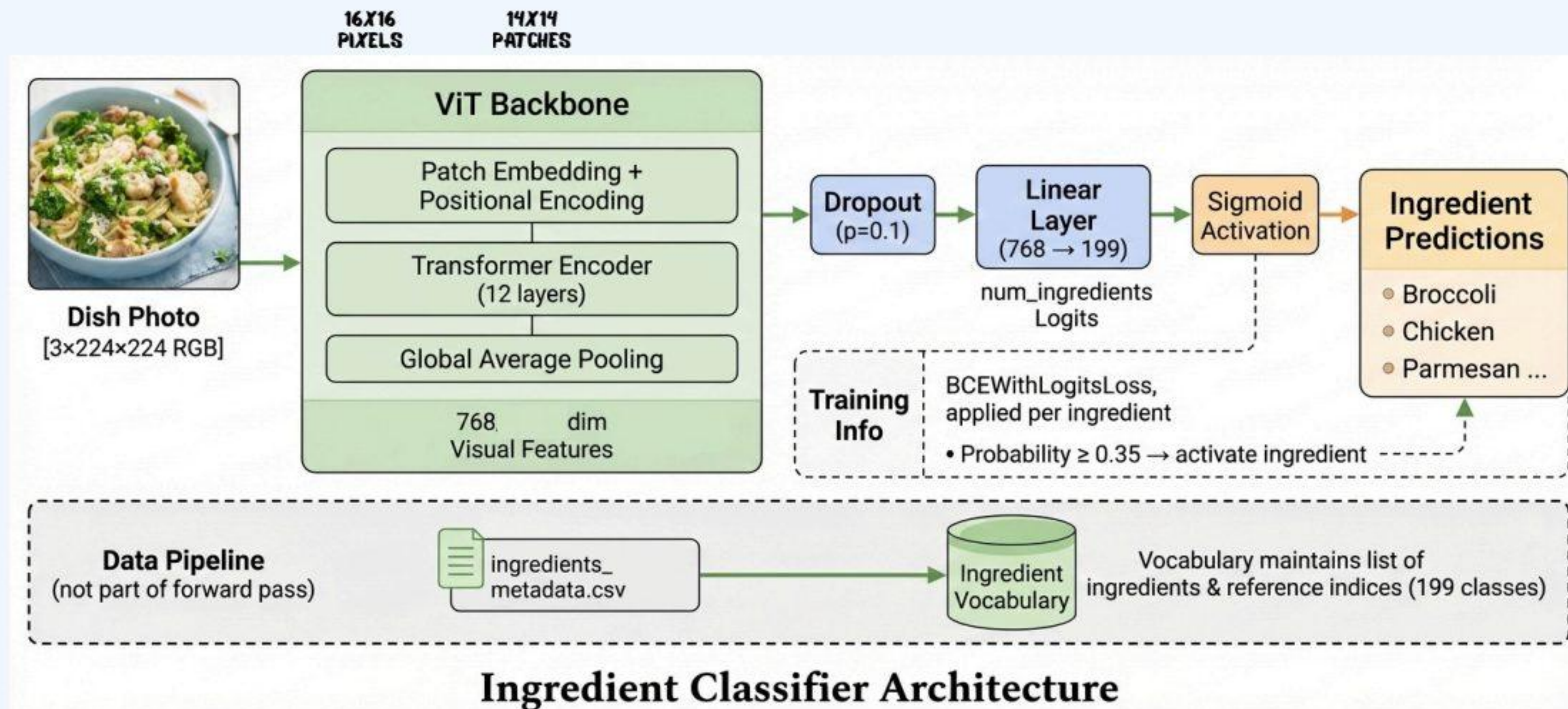
At a classification threshold of 0.2, the model achieved a precision of 0.20, recall of 0.80, and F1-score of 0.32 for ingredient detection.

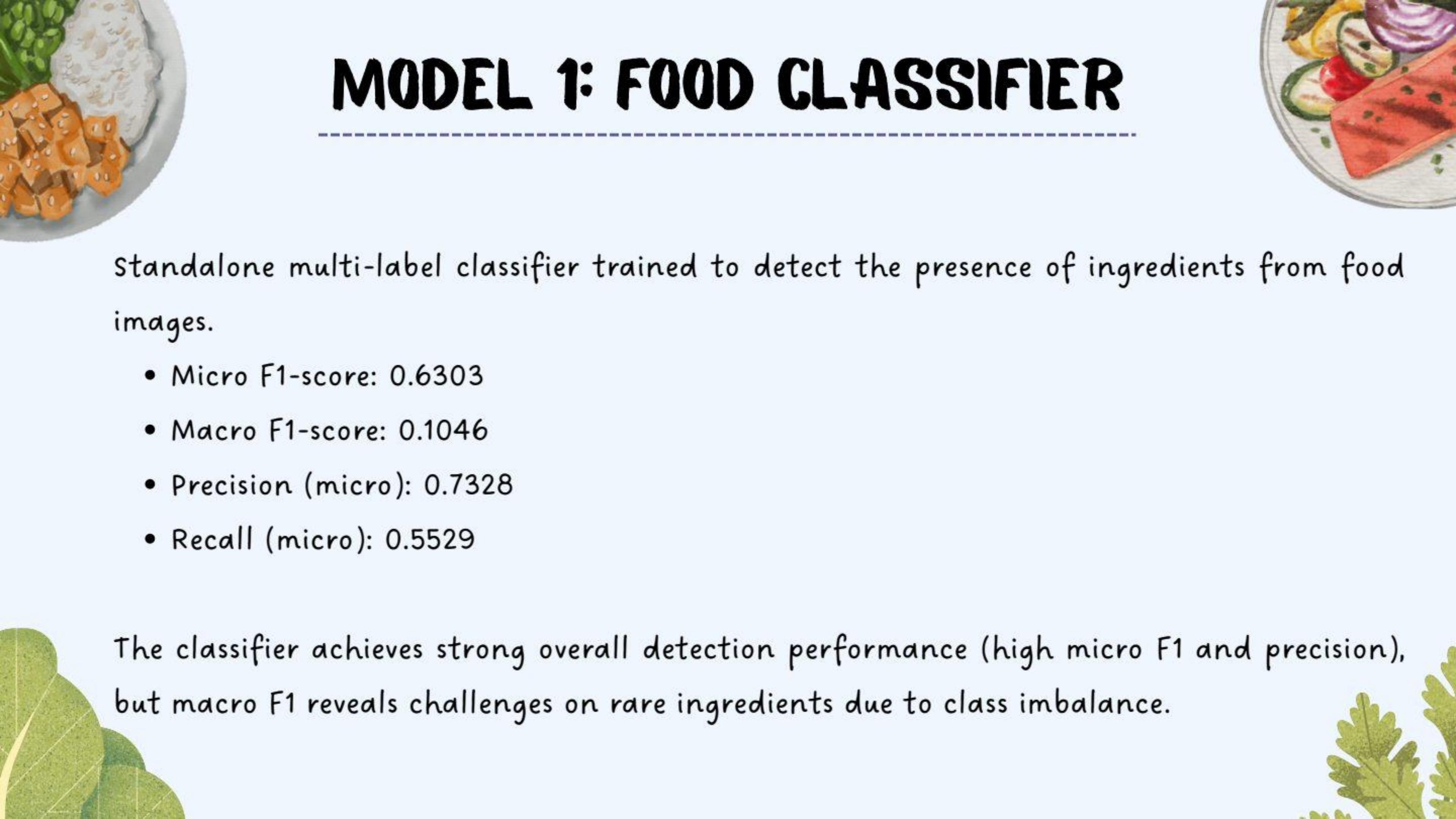
NUTRITIONAL REGRESSION PERFORMANCE

- Achieved a Mean Absolute Error (MAE) of 7.59g for carbohydrates.
- Loss curves of the baseline model showed efficient learning. Early stopping was triggered at epoch 59.



MODEL 1: FOOD CLASSIFIER



The slide features decorative food images in the corners. Top-left: A plate with white rice, green beans, and a salmon teriyaki bowl. Top-right: A plate with a salmon steak, cucumber, tomato, and onion. Bottom-left: A cluster of green leaves. Bottom-right: A sprig of green herbs.

MODEL 1: FOOD CLASSIFIER

Standalone multi-label classifier trained to detect the presence of ingredients from food images.

- Micro F1-score: 0.6303
- Macro F1-score: 0.1046
- Precision (micro): 0.7328
- Recall (micro): 0.5529

The classifier achieves strong overall detection performance (high micro F1 and precision), but macro F1 reveals challenges on rare ingredients due to class imbalance.



MODEL 2: FOOD CLASSIFIER + NNF (TRAINED SEPARATELY)

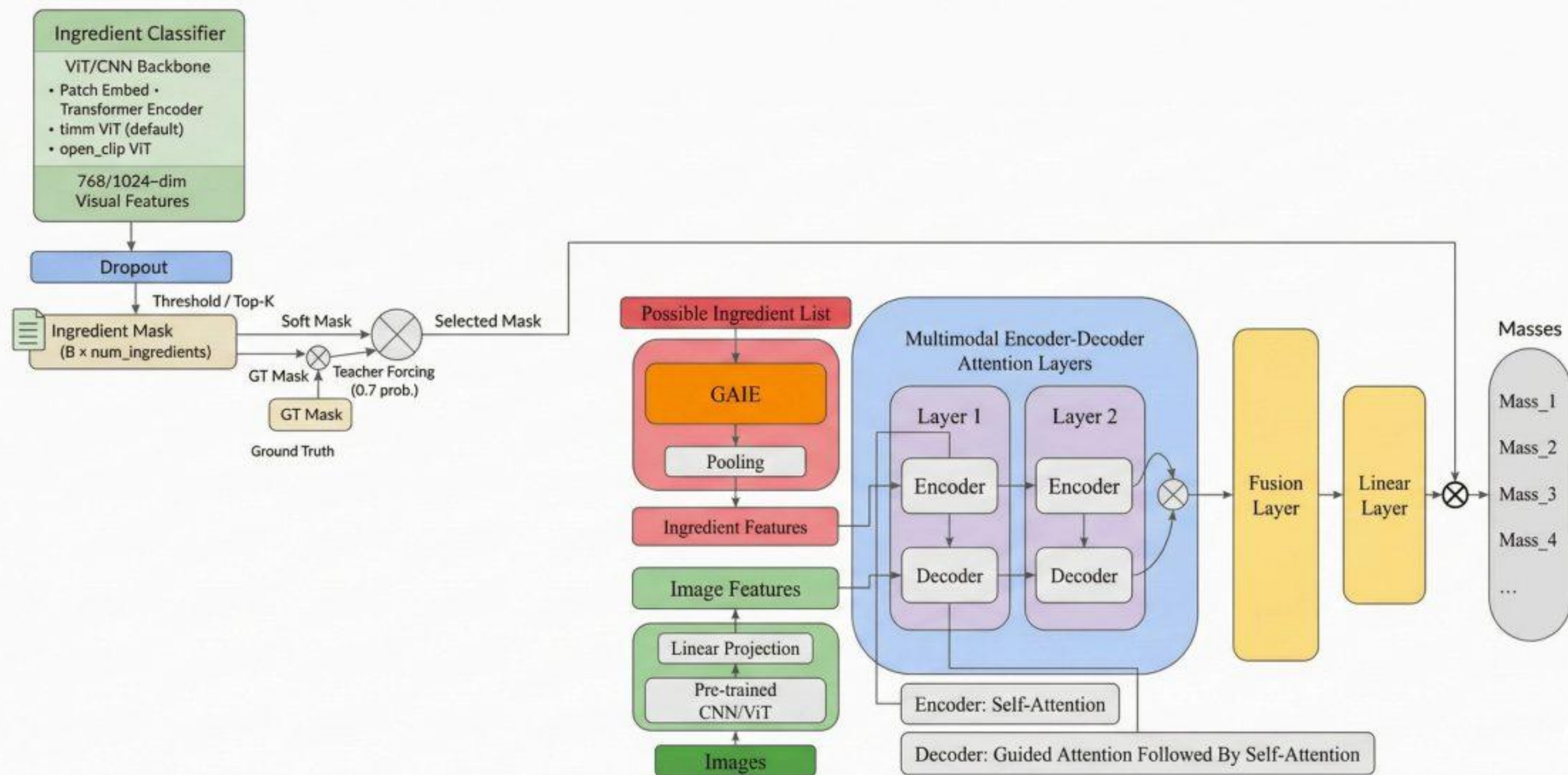
Two-stage pipeline where the classifier filters the ingredient list, and NutriFusionNet estimates ingredient masses. Models are trained independently.

- Micro F1-score: 0.6303
- Macro F1-score: 0.1046
- Precision: 0.7328
- Recall: 0.5529
- Carbohydrate MAE: 8.62 g

Filtering ingredients with the classifier improves interpretability and stabilizes predictions, but the mismatch between training and inference limits regression accuracy.



MODEL 3: FOOD CLASSIFIER + NUTRIFUSIONNET (JOINT MODEL)



The top of the slide features two circular inset images of food. The left image shows a plate with orange-colored cubes, possibly salmon or sweet potato, topped with a white sauce and green garnishes. The right image shows a plate with a piece of salmon, sliced cucumbers, tomatoes, and onions. The main title is centered between these images.

MODEL 3: FOOD CLASSIFIER + NUTRIFUSIONNET (JOINT MODEL)

Joint pipeline where the classifier guides NutriFusionNet during training and inference, reducing sparsity and improving alignment between ingredient detection and mass regression.

- Carbohydrate MAE: 5.3 g $\rightarrow$ Decreased by 30%

The joint model achieves the best overall performance, significantly improving both ingredient detection and nutritional estimation by reducing train-test mismatch and focusing regression on plausible ingredients.

A small, stylized green leaf illustration is located in the bottom right corner of the slide.

EXAMPLES

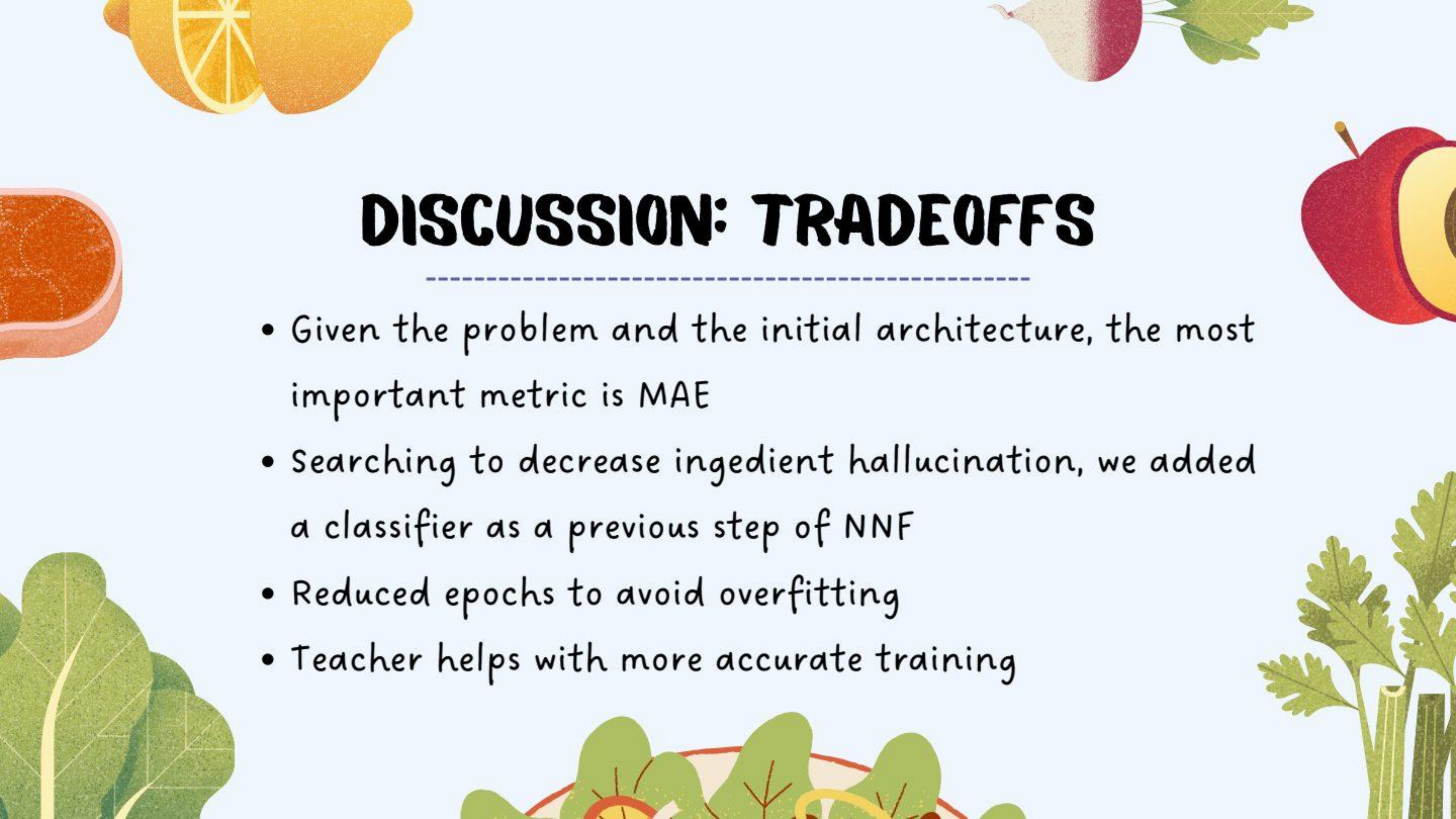
- Ingredients: ['garlic', 'pasta', 'broccoli', 'caesar salad', 'chicken']
- Estimated carbs (g): 31.27
- Real carbs (g): 40
- Carbs MAE: 8.73



EXAMPLES

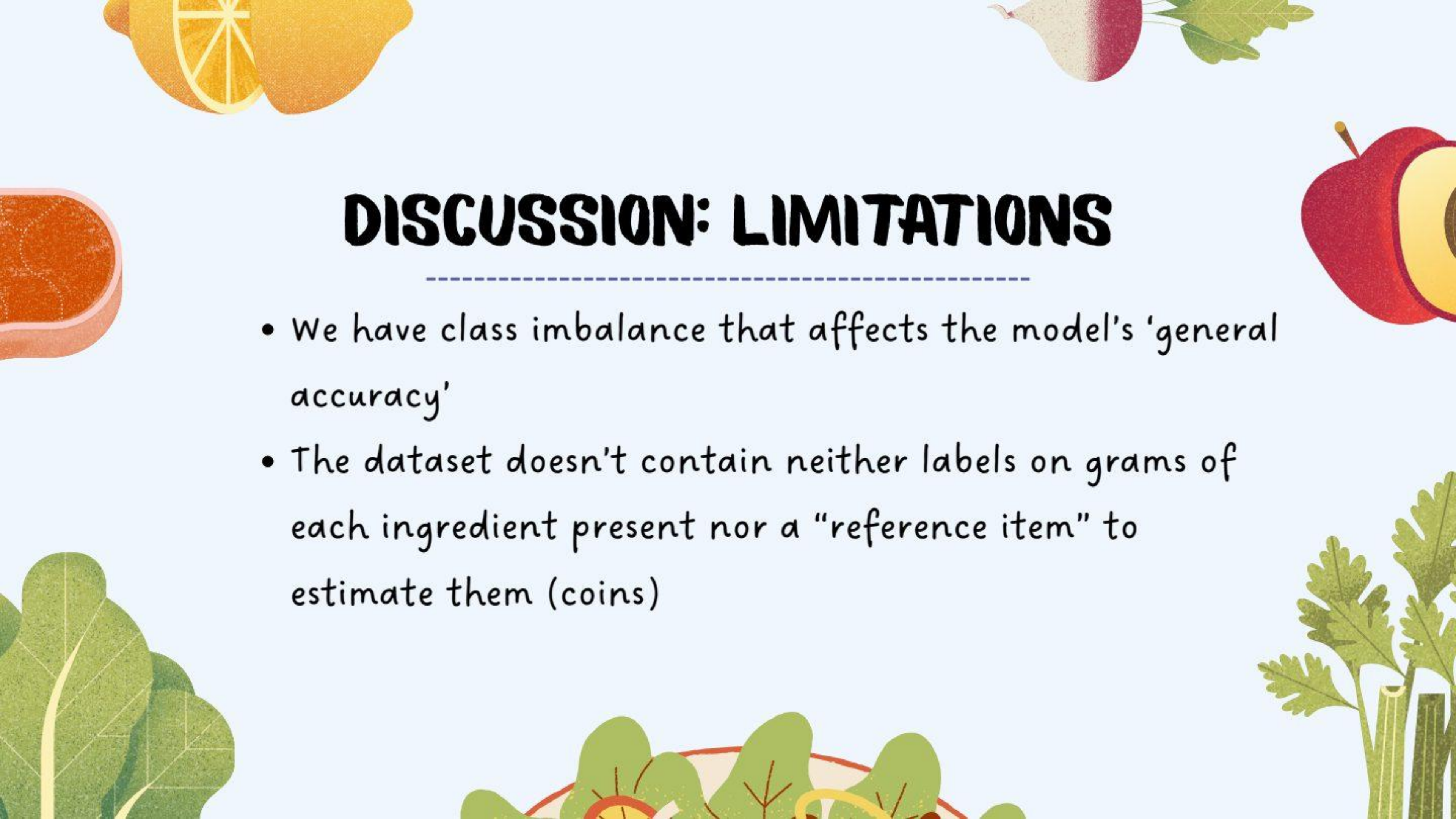
- Ingredients: ['garlic', 'onions', 'carrot', 'salt', 'olive oil']
- Estimated carbs (g): 46.77
- Real carbs (g): 70
- Carbs MAE: 23.23





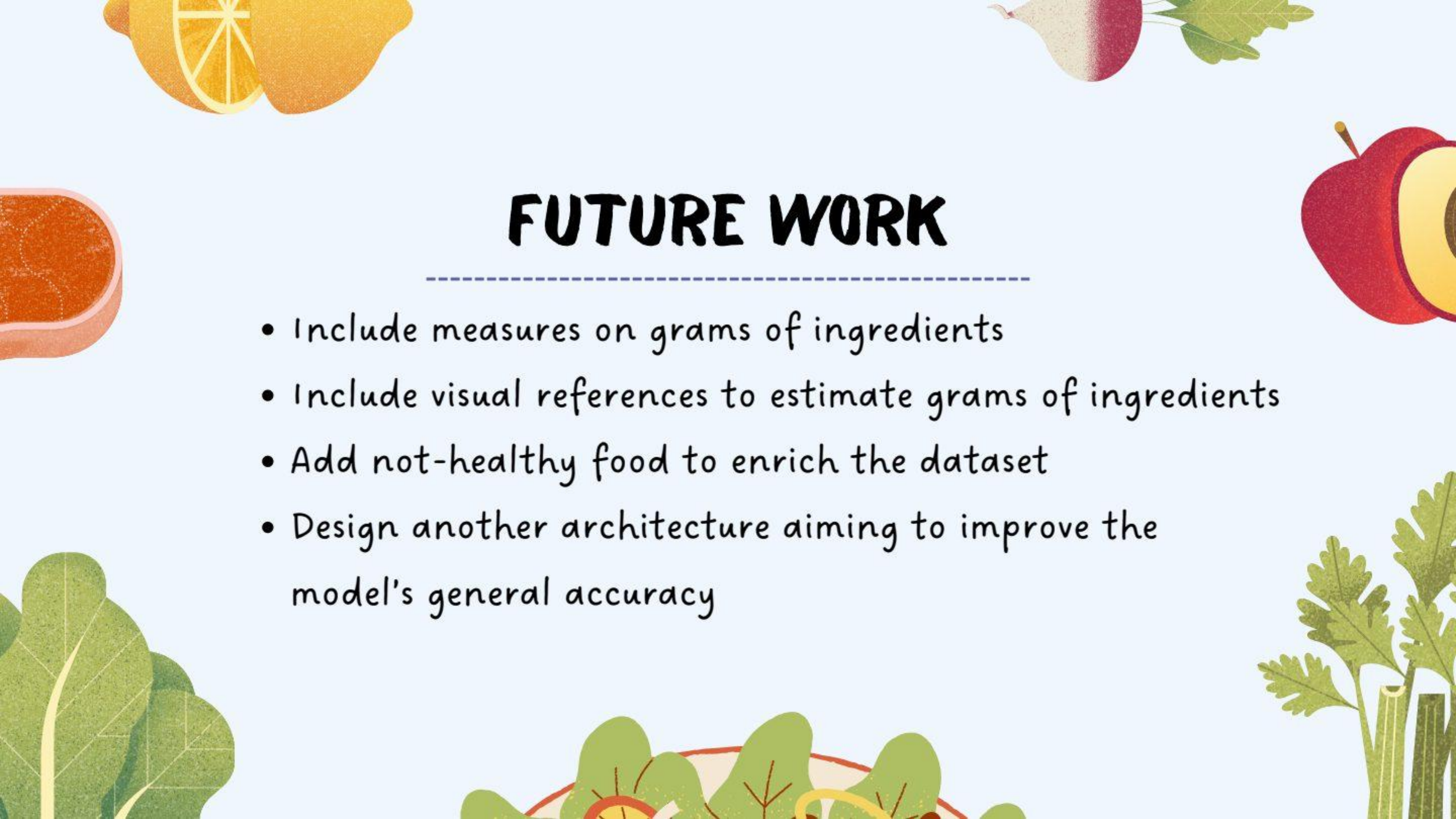
DISCUSSION: TRADEOFFS

- Given the problem and the initial architecture, the most important metric is MAE
- Searching to decrease ingredient hallucination, we added a classifier as a previous step of NNF
- Reduced epochs to avoid overfitting
- Teacher helps with more accurate training



DISCUSSION: LIMITATIONS

- We have class imbalance that affects the model's 'general accuracy'
- The dataset doesn't contain neither labels on grams of each ingredient present nor a "reference item" to estimate them (coins)

The slide features several stylized food illustrations: a lemon slice and whole lemon in the top left; a radish in the top right; a red apple with a bite taken out in the middle right; a piece of salmon in the middle left; a green leafy vegetable in the bottom left; and a bunch of green herbs in the bottom right.

FUTURE WORK

- Include measures on grams of ingredients
- Include visual references to estimate grams of ingredients
- Add not-healthy food to enrich the dataset
- Design another architecture aiming to improve the model's general accuracy